

MASCOT
GOES
HERE

KID INSTRUCTIONS

You've learned loops, conditionals, and nested loops. Now let's put them ALL together to make a real algorithm — one that real programmers use every single day!

Sort

Sort — To put things in order, like smallest to largest, A to Z, or shortest to tallest.

Example: When you line up by height in P.E. class, your teacher is sorting you — usually shortest to tallest!

How It Works — Bubble Sort

Bubble sort is one of the simplest sorting algorithms. Here are the rules:

1. Look at the first TWO items
2. Are they in the wrong order? SWAP them.
3. Move to the NEXT pair
4. Repeat until you reach the end — that's one PASS
5. Keep doing passes until nothing swaps. Now it's sorted!

Worked Example — Sort: [3] [1] [4] [2]

Start:

3

1

4

2

Pass 1: 3 < 1 SWAP, 3 < 4 keep, 4 < 2 SWAP → [1, 3, 2, 4]

Pass 2: 1 < 3 keep, 3 < 2 SWAP, 3 < 4 keep → [1, 2, 3, 4]

Pass 3: Check all pairs → no swaps needed → DONE!

Notice — the biggest number (4) **BUBBLED UP** to the end first. That's why it's called bubble sort!

Basic Version

Sort the Numbers — Use bubble sort to put these numbers in order from smallest to largest. Write the list after each pass.

Start:

5

2

8

3

Pass 1:

--

--

--

--

Pass 2:

--

--

--

--

Pass 3:

--

--

--

--

How many passes did it take?

Is the list sorted?

☐ YES ☐ NO

Challenge Version

PART A — Line Up the Friends

Imagine you're lining up 5 friends by height (shortest to tallest). Their heights in cm are:

Start:

140

125

150

130

145

Use bubble sort. Write each pass below. (You may not need all 4 rows — leave the last row empty if sorted.)

Pass 1:

--

--

--

--

--

Pass 2:

--

--

--

--

--

Pass 3:

--

--

--

--

--

Pass 4:

--

--

--

--

--

How many passes were needed?

→

PART B — Why Does Bubble Sort Need NESTED Loops?

Bubble sort has TWO loops working together — the OUTER loop counts passes, and the INNER loop compares pairs within a pass.

Repeat until no swaps happen:

Repeat for each pair in the list:

IF the pair is in the wrong order THEN swap them

Which loop runs more times in total — the outer loop or the inner loop? Why?

Self-Check

- ☐ I followed the worked example and understood why 4 "bubbled up" first
- ☐ I sorted the numbers [5, 2, 8, 3] using bubble sort
- ☐ I sorted the 5 heights and counted how many passes it took
- ☐ I can explain why bubble sort uses nested loops

Reflection Box

Bubble sort works, but is it FAST? If you had to sort 100 numbers, would bubble sort feel slow or quick? Why do you think real computers use FASTER sorting algorithms most of the time?

Bonus: Sort Real Things

Time to leave the page! Grab 5 things from around your house — books, toys, coins, anything! Line them up in a random order. Now use bubble sort to put them in order by size (smallest to biggest).

Items I sorted:

1. _____
2. _____
3. _____
4. _____
5. _____

What order did you sort them in? (size, weight, color, alphabet?)

→ _____

How many passes did you need? _____

Bonus reflection: Did doing it with REAL things feel different from doing it on paper?

Why?

TEACHER / PARENT NOTE

This is the first named algorithm in the workbook — and it uses every concept your child has learned so far: sequence (Activity 2), loops (Activities 3-4), nested loops (Activity 4), conditionals (Activity 5), and even debugging (Activity 7) when sorts go wrong. Bubble sort isn't the fastest sorting algorithm — but it's the most intuitive, which is why nearly every CS course on Earth starts here. Discussion prompt: Ask your child to spot something in real life that gets sorted regularly (a library shelf, a hand of cards, a classroom by height) and talk through what algorithm might be at work.